

MBEYA UNIVERSITY OF SCIENCE AND TECHNOLOGY
COLLEGE OF INFORMATION AND COMMUNICATION TECHNOLOGY

Test 01: ET 8208 Object-Oriented Programming

Date 2nd May 2024

Instructions

- Answer all questions
- The time allocated for this test is sixty (60) minutes

Question One

What do you understand by the following terms in object-oriented programming

- i. Class
- ii. Methods Overloading
- iii. Polymorphism
- iv. Encapsulation
- v. Access modifiers
- vi. Static members

[3 Marks]

Question two

- i. With at least one example from each, differentiate Procedural programming from Object-oriented Programming
- ii. Write a flow chart for a method to convert decimal numbers to binary numbers

[2 Marks]

[4 Marks]

Question three

- i. Using JAVA running codes, differentiate parameterized from non-parameterized constructors.
- ii. What are the outputs of the JAVA program below?

[3 Marks]

[3 Marks]

```
import java.io.*;
class Marks{
    int m1;
    int m2;
}
class MarksDemo
{
    public static void main(String args[])
    { //first object
        Marks obj1 = new Marks();
        obj1.m1 = 50;
        obj1.m2 = 80;
        //second object
        Marks obj2 = new Marks();
        obj2.m1 = 80;
        obj2.m2 = 60;
        //displaying marks for first object
        System.out.println("Marks for first object:");
        System.out.println(obj1.m1);
        System.out.println(obj1.m2);
        //displaying marks for second object
        System.out.println("Marks for second object:");
        System.out.println(obj2.m1);
        System.out.println(obj2.m2);
    }
}
```